Sweetness is one of the five basic tastes known to be sensed by humans. In a research study of sweet taste perception, individuals drank solutions containing only one of two monosaccharides (Figure 1) or two disaccharides (Figure 2) on different days and were asked to rate the relative sweetness of each solution.

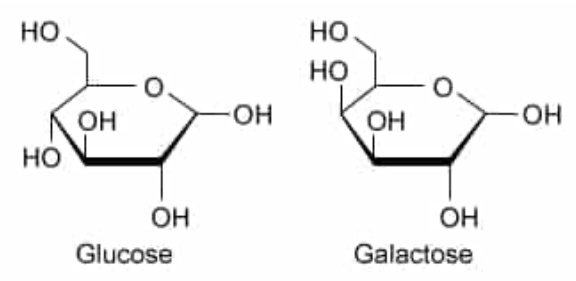


Figure 1 Structures of the monosaccharides studied

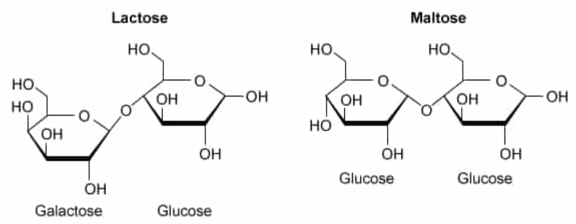


Figure 2 Structures of the disaccharides studied

The sweetness and sugar concentration of the monosaccharide and disaccharide solutions are presented in Table 1. The solutions were prepared the night before each testing day to allow for complete dissolution of the sugars.

Table 1 Concentration and Sweetness of Sugar Solutions

Sugar solution	Concentration (M)	Sweetness
Lactose (galactose-glucose dimer)	1	0.2
Maltose (glucose-glucose dimer)	1	0.4
Glucose	1	0.5
Galactose	1	0.5

In a second experiment, the measurements were repeated with the addition of glucosidase enzymes to each solution during preparation. Tasting occurred the next day as in the first experiment, thereby allowing glucosidase cleavage of all glycosidic bonds.

Of the four original solutions studied, which of the following pairs of solutions contain epimers?

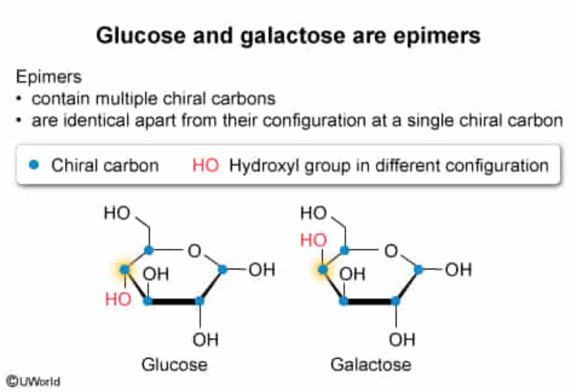
I. Maltose and lactose
II. Galactose and glucose
III. Maltose and glucose

IV. Lactose and galactose

- A. I only [10%]
- B. II only [75%]
- C. I and III only [4%]
- D. II and IV only [9%]

Correct
76% answered correctly

Explanation:



The function of a biological molecule is determined by its structure and its environment. Thus, classification of biomolecules according to the structural features they possess can often aid in the identification of biomolecules that share certain properties or functional roles. Nearly all carbohydrates contain one or more carbon atoms that are part of a chiral center, and this characteristic can be used as one basis for classification.

The passage describes a study in which the sweetness of various sugar solutions is determined, and the question asks which pairs of solutions contain sugars that are epimers. **Epimers** are molecules that contain **multiple chiral carbons** and are **identical apart from** their **configuration at** a **single chiral carbon**. Of the sugar pairs provided, only glucose and galactose meet both these requirements **(Number II)**.

(Number I) Maltose and lactose are disaccharides composed of different sugar monomers (Figure 1). The configurations of the chiral centers at carbons 1 and 4 of the galactose in lactose are different from the configurations of the chiral centers at carbons 1 and 4 of the glucose at the nonreducing end of maltose. Therefore, lactose and maltose are not epimers.

(Number III) Although maltose is a disaccharide formed from two glucose monomers, the structures of maltose and glucose differ in more respects than just the configuration around a single carbon. Therefore, maltose and glucose are not epimers.

(Number IV) Although lactose is formed from galactose and glucose monomers, which are epimers, lactose is a disaccharide and galactose is a monosaccharide. Therefore, the structures of lactose and galactose differ in more respects than just the configuration around a single carbon, and cannot be epimers.

Educational objective:
Epimers contain more than one chiral carbon and are identical apart from their configuration at a single chiral carbon.

Time spent: 0 sec	**QID**:403550
Subject: Biochemistry	**System:** Carbohydrates, Nucleotides, and Lipids

Sweetness is one of the five basic tastes known to be sensed by humans. In a research study of sweet taste perception, individuals drank solutions containing only one of two monosaccharides (Figure 1) or two disaccharides (Figure 2) on different days and were asked to rate the relative sweetness of each solution.

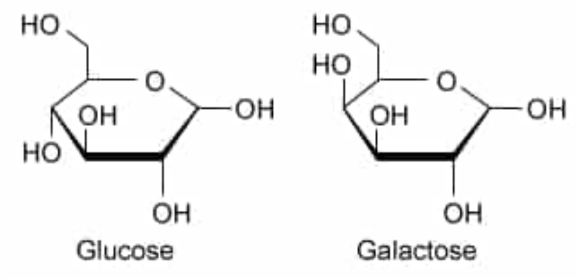


Figure 1 Structures of the monosaccharides studied

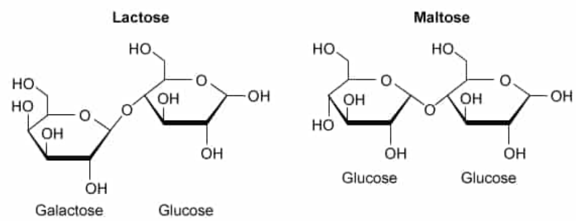


Figure 2 Structures of the disaccharides studied

The sweetness and sugar concentration of the monosaccharide and disaccharide solutions are presented in Table 1. The solutions were prepared the night before each testing day to allow for complete dissolution of the sugars.

Table 1 Concentration and Sweetness of Sugar Solutions

Sugar solution	Concentration (M)	Sweetness
Lactose (galactose-glucose dimer)	1	0.2
Maltose (glucose-glucose dimer)	1	0.4
Glucose	1	0.5
Galactose	1	0.5

In a second experiment, the measurements were repeated with the addition of glucosidase enzymes to each solution during preparation. Tasting occurred the next day as in the first experiment, thereby allowing glucosidase cleavage of all glycosidic bonds.

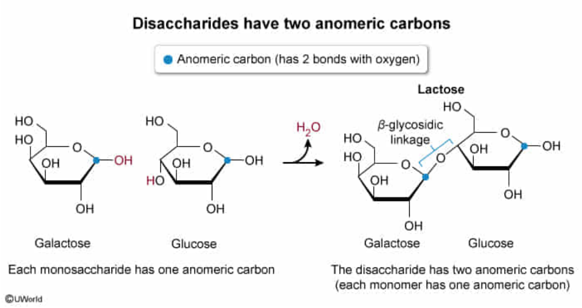
What is the total number of anomeric carbons in disaccharides that are linked by β-glycosidic bonds, such as lactose?

- A. 0 [3%]
- B. 1 [40%]
- C. 2 [54%]

○ D. 10 [2%]

Correct
53% answered correctly

Explanation:



Monosaccharides containing four or more carbons often exist primarily as ring structures formed when the carbonyl group reacts with a hydroxyl group (eg, as in the formation of ring forms of glucose). In both the linear and the ring forms, the number of carbons in the sugar and the specific arrangement of constituents bound to each carbon determine the identities of the sugar molecule (eg, glucose, galactose) as well as the carbons themselves (eg, C1, C2).

The passage describes a study of the sweetness of sugar-containing solutions, including those containing the disaccharides lactose and maltose, and the question asks for the total number of anomeric carbons in disaccharides linked by β-glycosidic bonds. The anomeric carbon in a monosaccharide is the carbon that has **two bonds with oxygen**. **Each sugar residue** in a disaccharide or polysaccharide **has one anomeric carbon**. Therefore, all **disaccharides** (regardless of configuration) **have two anomeric carbons**.

(Choices A and B) Some disaccharides, such as sucrose lack a *free* anomeric carbon (ie, an anomeric carbon that is not involved in a glycosidic bond), and others have only one *free* anomeric carbon. However, the *total* number of anomeric carbons (including those involved in glycosidic bonds) is always two for disaccharides, three for trisaccharides, and so on.

(Choice D) The total number of chiral carbons in the ring forms of glucose and galactose, the two monosaccharides that are linked in lactose, is ten. However, this fact is unrelated to the number of anomeric carbons in lactose.

Educational objective:
In a monosaccharide, the anomeric carbon is the one that has two bonds with oxygen. Because each monosaccharide has one anomeric carbon, disaccharides have two anomeric carbons.

Time spent:0 sec
Subject:
Biochemistry

QID:403551
System:
Carbohydrates, Nucleotides, and Lipids

Sweetness is one of the five basic tastes known to be sensed by humans. In a research study of sweet taste perception, individuals drank solutions containing only one of two monosaccharides (Figure 1) or two disaccharides (Figure 2) on different days and were asked to rate the relative sweetness of each solution.

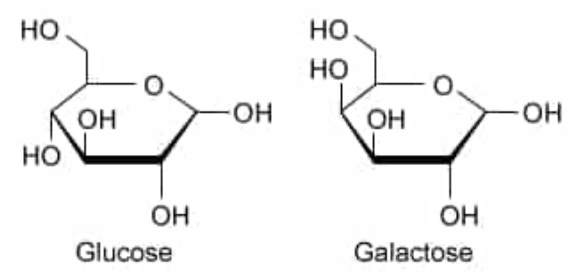


Figure 1 Structures of the monosaccharides studied

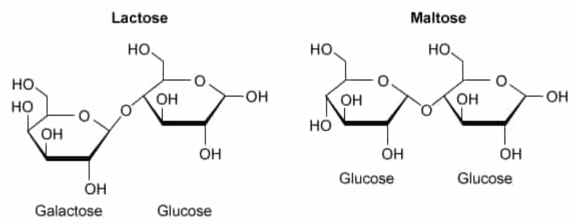


Figure 2 Structures of the disaccharides studied

The sweetness and sugar concentration of the monosaccharide and disaccharide solutions are presented in Table 1. The solutions were prepared the night before each testing day to allow for complete dissolution of the sugars.

Table 1 Concentration and Sweetness of Sugar Solutions

Sugar solution	Concentration (M)	Sweetness
Lactose (galactose-glucose dimer)	1	0.2
Maltose (glucose-glucose dimer)	1	0.4
Glucose	1	0.5
Galactose	1	0.5

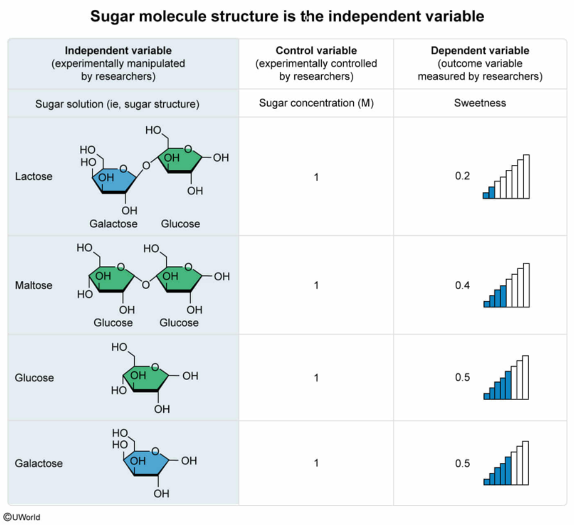
In a second experiment, the measurements were repeated with the addition of glucosidase enzymes to each solution during preparation. Tasting occurred the next day as in the first experiment, thereby allowing glucosidase cleavage of all glycosidic bonds.

What was the independent variable in the first study described in the passage?

- A. Concentration of the sugar solutions [15%]
- ✔ B. Structure of the sugar molecules in the solutions [74%]
- C. Sweetness of the sugar solutions [7%]
- D. Order in which the sugar solutions were tasted [2%]

Correct
74% answered correctly

Explanation:



Disaccharides (eg, maltose, lactose) are formed by the linkage of two monosaccharides via a glycosidic bond. The passage describes an experiment in which the sweetness of monosaccharide and disaccharide solutions was studied, and the question asks for the independent variable in the experiment.

In research studies, **independent variables** are those **varied through manipulation** (eg, by assigning treatments) **or selection** (eg, by sampling at specific times) **by** the **researchers**. Dependent variables are those variables that change in response to variation in the independent variable(s).

The **researchers** experimentally **varied** the **structure** of the sugar molecules in the experiment by choosing the monosaccharides and disaccharides to study (independent variable) **and measured** the **response of sweetness** to variation in structure (dependent variable). Therefore, the **structure of** the **sugars** in the solutions **was** the **independent variable** in the experiment.

(Choice A) Sugar solution concentration *can* be an independent variable if it is varied in the experiment, but in this case concentration is *not* the independent variable because it was controlled rather than varied by the researchers. In a *different* study, the researchers might choose to study only one type of sugar (thereby making sugar structure a *control* variable) and vary the sugar concentration (thereby making sugar concentration the *independent* variable).

(Choice C) Sweetness is the dependent variable in the experiment (ie, the variable that changed in response to the variation in the independent variable).

(Choice D) Tasting order is a potential independent variable (or control variable). However, because the specific order was not mentioned in the description of the experiment, tasting order can be excluded as the independent variable of this specific study.

Educational objective:
Independent variables are those experimentally varied by researchers to determine if they influence dependent variables.

Time spent: 0 sec	**QID**:403552
Subject: Biochemistry	**System:** Carbohydrates, Nucleotides, and Lipids

Sweetness is one of the five basic tastes known to be sensed by humans. In a research study of sweet taste perception, individuals drank solutions containing only one of two monosaccharides (Figure 1) or two disaccharides (Figure 2) on different days and were asked to rate the relative sweetness of each solution.

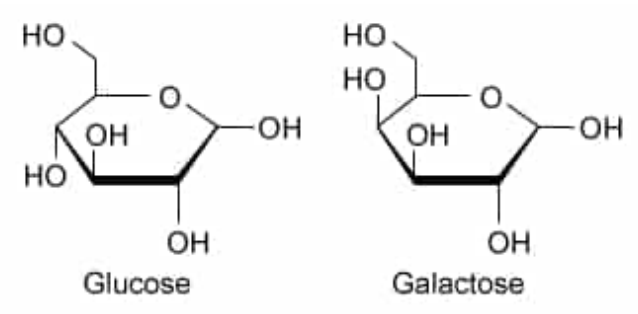


Figure 1 Structures of the monosaccharides studied

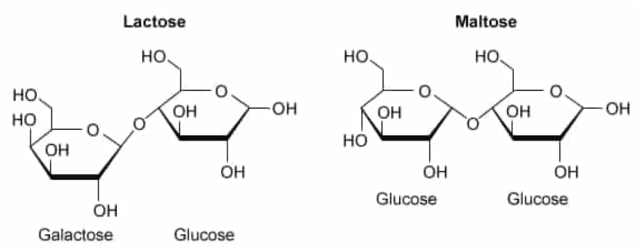


Figure 2 Structures of the disaccharides studied

The sweetness and sugar concentration of the monosaccharide and disaccharide solutions are presented in Table 1. The solutions were prepared the night before each testing day to allow for complete dissolution of the sugars.

Table 1 Concentration and Sweetness of Sugar Solutions

Sugar solution	Concentration (M)	Sweetness
Lactose (galactose-glucose dimer)	1	0.2
Maltose (glucose-glucose dimer)	1	0.4
Glucose	1	0.5
Galactose	1	0.5

In a second experiment, the measurements were repeated with the addition of glucosidase enzymes to each solution during preparation. Tasting occurred the next day as in the first experiment, thereby allowing glucosidase cleavage of all glycosidic bonds.

If the researchers used the form of galactose most common in nature (ie, d-galactose), the sequence of stereochemical configurations from carbon 2 to carbon 5 must be:

A. *S*, *R*, *R*, *S* [22%]

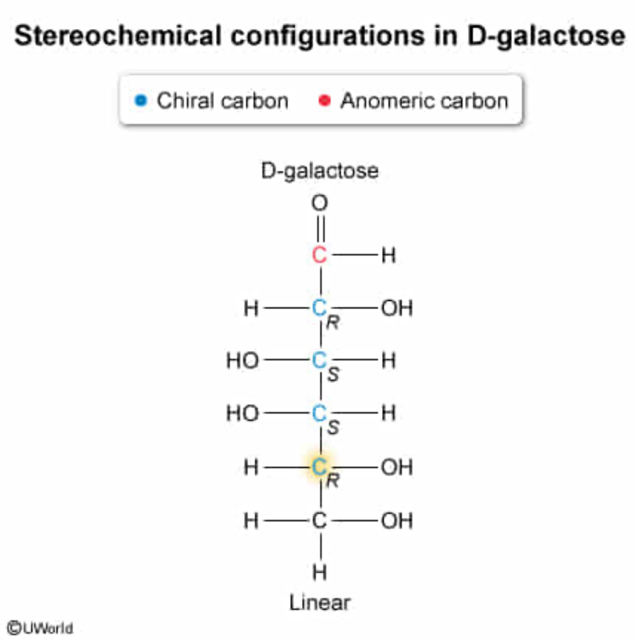
B. *R*, *S*, *R*, *S* [16%]

✔ C. *R*, *S*, *S*, *R* [44%]

D. *R*, *R*, *R*, *S* [16%]

Correct
43% answered correctly

Explanation:



Stereocenters are atoms bound to four different groups in a molecule. Nearly all sugars contain stereocenters, and each stereocenter in a sugar can be classified as having an *R* or *S* configuration according to the arrangement of these groups about the central atom. Whereas *individual stereocenters* can be classified as *R* or *S*, *complete molecules* containing stereocenters can be classified as having an L or D configuration, based on the configuration at the stereocenter farthest from the anomeric carbon.

Most sugars in nature are of the D configuration rather than the L configuration. Because the L and D forms of a sugar are mirror images, each of their stereocenters will be of opposite configuration.

The question asks for the sequence of stereocenter configurations in the most common naturally occurring form of galactose (ie, D-galactose). For a **sugar** to be **of** the **D form**, the **stereocenter farthest from** the **anomeric carbon must**

have the ***R* configuration** (note that this only applies to sugars and not to all chiral molecules). Therefore, the sequence of *R*, *S*, *S*, *R* stereocenter configurations is the only one that could possibly be that of d-galactose.

(Choices A, B, and D) Because the L or D designation is based solely on the configuration at the stereocenter *farthest* from the anomeric carbon, the configuration at carbon 5 must be *R* (*not S*). Notably, the *S*, *R*, *R*, *S* sequence of stereocenter configurations corresponds to *L-galactose*, the enantiomer of D-galactose.

Educational objective:

D- and L-sugars have opposite *R* and *S* configurations at each stereocenter. The stereocenter farthest from the anomeric carbon is used to classify the configuration of a sugar, and will be in the *R* configuration for a D-sugar and in the *S* configuration for an L-sugar.

Time spent:0 sec	**QID**:403553
Subject:	**System:**
Biochemistry	Carbohydrates, Nucleotides, and Lipids

Sweetness is one of the five basic tastes known to be sensed by humans. In a research study of sweet taste perception, individuals drank solutions containing only one of two monosaccharides (Figure 1) or two disaccharides (Figure 2) on different days and were asked to rate the relative sweetness of each solution.

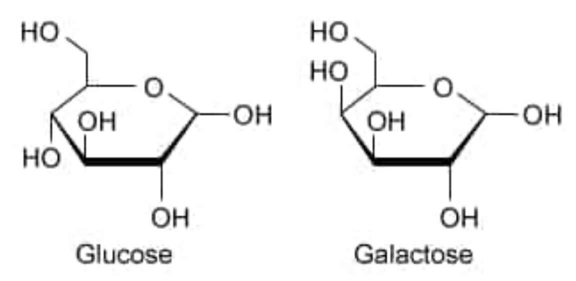


Figure 1 Structures of the monosaccharides studied

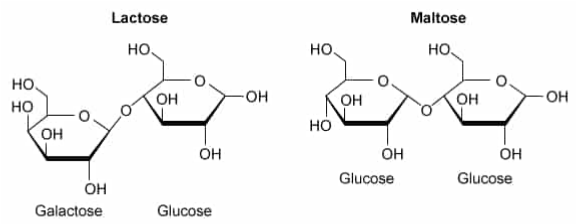


Figure 2 Structures of the disaccharides studied

The sweetness and sugar concentration of the monosaccharide and disaccharide solutions are presented in Table 1. The solutions were prepared the night before each testing day to allow for complete dissolution of the sugars.

Table 1 Concentration and Sweetness of Sugar Solutions

Sugar solution	Concentration (M)	Sweetness
Lactose (galactose-glucose dimer)	1	0.2
Maltose (glucose-glucose dimer)	1	0.4
Glucose	1	0.5
Galactose	1	0.5

In a second experiment, the measurements were repeated with the addition of glucosidase enzymes to each solution during preparation. Tasting occurred the next day as in the first experiment, thereby allowing glucosidase cleavage of all glycosidic bonds.

What was the purpose of the measurements performed after glucosidase treatment?

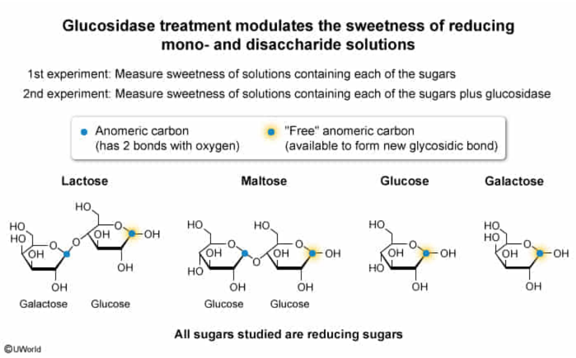
- ◯ A. To determine whether glucosidase treatment can modulate the sweetness of nonreducing sugar solutions, as established in the first experiment [24%]
- ✔ ◉ B. To determine whether glucosidase treatment can modulate the sweetness of reducing sugar solutions, as established in the first experiment [67%]

○ C. To determine whether the glucosidase concentration is linearly related to the magnitude of its effect on the sweetness of sugar solutions, as established in the first experiment [6%]

○ D. To determine whether glucosidase is more or less sweet than the monosaccharide and disaccharide solutions studied in the first experiment [2%]

Correct
66% answered correctly

Explanation:



All sugars contain anomeric carbons. Because anomeric carbons take part in glycosidic bonding, at least one anomeric carbon per monosaccharide residue will be bound to another sugar in disaccharides and polysaccharides. **Sugars with** a **free anomeric carbon** are **called reducing sugars** whereas sugars **without** a **free anomeric carbon** are called **nonreducing sugars**. Because they lack a free anomeric carbon, nonreducing sugars cannot be lengthened by glycosidic bonding.

The passage describes measurements of sugar solution sweetness, initially in the absence of glucosidase treatment and, in a second set of measurements, after glucosidase treatment. **Each of** the **solutions** in the study **contains** a **reducing sugar**, and the question asks for the purpose of the second set of measurements with glucosidase treatment. Therefore, the **second set** of **measurements** allows the researchers to **determine whether glucosidase** treatment **alters** the **sweetness of** the **reducing sugar solutions** as established in the first experiment.

(Choice A) The sugars studied by the researchers are all reducing (*not* nonreducing) sugars.

(Choice C) To determine whether the glucosidase concentration is linearly related to the magnitude of its effect on the sweetness of sugar solutions as measured in the first experiment, the researchers would need to use several glucosidase concentrations (*not* just one concentration).

(Choice D) If the researchers wished to compare the sweetness of glucosidase to that of the sugar solutions, the most direct approach would be to measure the sweetness of a solution containing only glucosidase (*not* to add glucosidase to each of the sugar solutions). Therefore, this is not the purpose of the measurements performed after treating the solutions with glucosidase.

Educational objective:
Reducing sugars are those with a free anomeric carbon (ie, one not involved in a glycosidic linkage).

Time spent: 0 sec	**QID**:403554
Subject:	**System:**
Biochemistry	Carbohydrates, Nucleotides, and Lipids

Sweetness is one of the five basic tastes known to be sensed by humans. In a research study of sweet taste perception, individuals drank solutions containing only one of two monosaccharides (Figure 1) or two disaccharides (Figure 2) on different days and were asked to rate the relative sweetness of each solution.

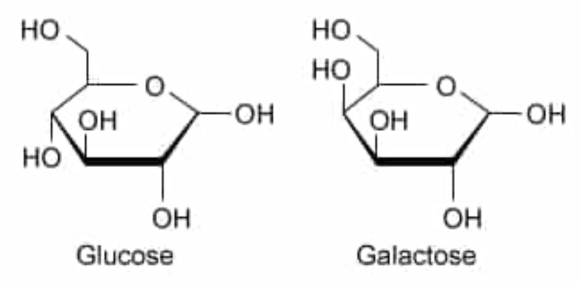


Figure 1 Structures of the monosaccharides studied

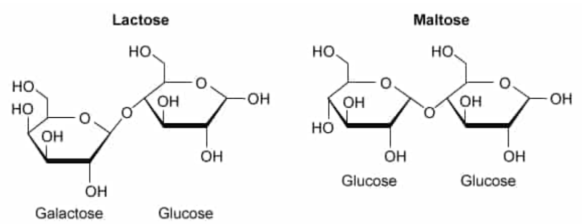


Figure 2 Structures of the disaccharides studied

The sweetness and sugar concentration of the monosaccharide and disaccharide solutions are presented in Table 1. The solutions were prepared the night before each testing day to allow for complete dissolution of the sugars.

Table 1 Concentration and Sweetness of Sugar Solutions

Sugar solution	Concentration (M)	Sweetness
Lactose (galactose-glucose dimer)	1	0.2
Maltose (glucose-glucose dimer)	1	0.4
Glucose	1	0.5
Galactose	1	0.5

In a second experiment, the measurements were repeated with the addition of glucosidase enzymes to each solution during preparation. Tasting occurred the next day as in the first experiment, thereby allowing glucosidase cleavage of all glycosidic bonds.

Which solution(s) will likely be rated as sweetest after the addition of glucosidase enzymes to each solution? (Note: Assume solution sweetness is determined equally by intrinsic sugar sweetness and sugar concentration.)

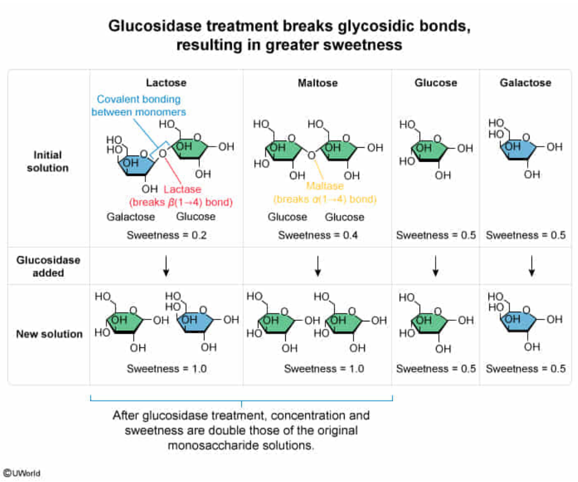
- A. Lactose only [4%]
- B. Maltose only [19%]

◯ C. Glucose and galactose equally [26%]

✔ ◉ D. Lactose and maltose equally [49%]

Correct
48% answered correctly

Explanation:



The class of molecules known as carbohydrates includes both simple **monomeric sugars** (eg, **glucose, galactose**) and **polymers** (ie, disaccharides or polysaccharides) composed of **linked sugar monomers**. The numerous roles played by carbohydrate polymers are related to the characteristics of their constituent sugar monomers and the types of linkages connecting them.

Some carbohydrate molecules activate specific plasma membrane receptors that give an individual the ability to taste sweetness. As depicted in the question table, sweetness varies according to the structure of carbohydrate monomers and polymers.

The question describes a study of sugar solution sweetness and asks for the solution(s) likely to be rated as sweetest if the experiment is repeated with the addition of glucosidase enzymes to the solutions. **Glucosidases** are enzymes that **break** the **glycosidic bonds** in sugar polymers. Glucosidase treatment would double the sugar concentrations of the disaccharide solutions by converting them to monomers whereas the sugar concentration in the original sugar monomer solutions would be unaffected by glucosidase treatment. Because the **total concentrations** of **glucose** and/or **galactose** (two equally sweet monosaccharides) would be **twice as high in** the **disaccharide solutions as** in the **monosaccharide solutions**, the **lactose** and **maltose solutions** would be **sweetest** after glucosidase treatment.

(Choices A and B) The passage states that overnight treatment with glucosidase will cleave the glycosidic bonds linking the monomers in maltose and lactose. The monosaccharides resulting from glucosidase treatment (ie, glucose and galactose) are equally sweet. Therefore, glucosidase-treated maltose and lactose solutions will be equally sweet.

(Choice C) Glucose and galactose are monosaccharides. Because they do not contain glycosidic linkages, neither their structure nor their sweetness will be affected by glucosidase treatment.

Educational objective:
Glucosidases are enzymes that facilitate the breaking of glycosidic bonds in sugar polymers (ie, disaccharides and polysaccharides).

Time spent: 0 sec	**QID:** 403555
Subject: Biochemistry	**System:** Carbohydrates, Nucleotides, and Lipids